The Debye–Boltzmann Initial Condition

A walkthrough for the electrochemistry-light reader

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PNP–BV forward solver, May 2026

Abstract

This note explains, from electrochemistry first principles, what the `debye_boltzmann` initial condition does, why it exists, and where each formula comes from. It is meant to be read end-to-end without prior PNP/electrochemistry background. Equations are given where they help, but the goal is to build the picture, not to be exhaustive. Code anchor: `Forward/bv_solver/forms_logc.py:581--952`.

Contents

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1 What problem is the IC actually solving?

The forward solver is a Firedrake finite-element discretization of a coupled nonlinear PDE system:

- One Nernst–Planck (NP) equation per dynamic species (O_2, H_2O_2, H^+),
- One Poisson equation for the electrostatic potential φ,
- Two Butler–Volmer (BV) reaction-rate boundary conditions on the electrode surface (R_1: $O_2 \rightarrow H_2O_2$, R_2: $H_2O_2 \rightarrow H_2O$).

Steady-state is reached by pseudo-time-stepping with backward Euler plus Newton's method (SNES). Newton needs a starting guess. That starting guess is the *initial condition* ("IC") discussed here.

A point of vocabulary

"IC" here is overloaded with the time-evolution sense of initial condition. We do not care about transient dynamics — the only thing we want from the IC is for it to be *close enough to the steady-state solution that Newton converges.* A bad IC means Newton diverges or stalls. A good IC means Newton converges in a handful of iterations. That is the entire point.

1.1 Why a simple IC fails

The naive IC — "set every concentration to its bulk value, set φ to a linear ramp from φ_{applied} at the electrode to 0 in the bulk" — works at low applied voltage but fails badly at moderate to high voltage. To see why, you need to understand what happens physically near the electrode.

2 Two-minute electrochemistry primer

2.1 The setup

Picture a 1D domain. Coordinate $y \in [0, 1]$ (nondimensionalized).

- $y = 0$ is the **electrode** (a metal disk you applied a voltage to).
- $y = 1$ is the **bulk** (well-mixed electrolyte far from the electrode).
- In between is the electrolyte: water with dissolved H^+, ClO_4^- (counterion), O_2, and H_2O_2.

When you hold the electrode at potential φ_{applied}, two things happen at once:

1. Charges **rearrange** to screen the field — positive ions are pushed away from a positive electrode, negative ions are pulled in (and vice-versa). This is purely electrostatic and happens essentially instantaneously.

2. BV **reactions** consume O_2 and H^+ and produce H_2O_2, then consume H_2O_2 and H^+ to make H_2O. This sets up diffusion gradients away from the electrode.

These two processes happen on *wildly different length scales.*

2.2 The Debye length: where screening lives

The screening length is the *Debye length*:

$$\lambda_D = \sqrt{\frac{\varepsilon V_T}{2F^2 c_b}}, \tag{1}$$

where ε is the electrolyte permittivity, $V_T = RT/F \approx 25.7$ mV at room temperature, F is the Faraday constant, and c_b is the bulk ion concentration.

For pH 4 with a 0.1 M counterion, $\lambda_D \sim 1$ nm. The full domain might be 100 µm thick. So the screening region is 10^{-4} of the domain. The PNP equations are stiff in exactly the sense that interesting physics is concentrated in a tiny corner.

2.3 The Boltzmann distribution: charge follows potential

Inside the Debye layer, ions reach a local equilibrium: their electrochemical potential becomes constant. For species i with charge z_i,

$$\mu_i = \ln c_i + z_i\varphi = \text{const} \implies \boxed{c_i(y) = c_i^{\text{bulk}} \exp\bigl(-z_i\,\psi(y)\bigr)} \tag{2}$$

where $\psi(y) = \varphi(y) - \varphi^{\text{bulk}}$ is the deviation from bulk potential, in units of V_T. (All φ quantities in this document are nondimensionalized by V_T, so factors of V_T vanish.)

This is the **Boltzmann distribution**. It is the equilibrium profile a single charged species takes in an electric field. *The entire IC machinery is built around this formula.*

Numerical concern. If the electrode is held at $\psi_D \equiv \varphi_{\text{electrode}} - \varphi^{\text{bulk}} = +30$, then for the proton ($z_H = +1$),

$$c_H^{\text{surface}} = c_H^{\text{bulk}} \cdot e^{-30} \approx c_H^{\text{bulk}} \cdot 10^{-13}.$$

Thirteen orders of magnitude. A Newton solver starting from $c_H = c_H^{\text{bulk}}$ everywhere has to discover this depletion in one step, which it cannot.

2.4 Outer vs. inner: two regions, two physics

Combining the two effects above gives a clean two-region picture:

The two-region picture

Outer region ($y \gtrsim \lambda_D$, the bulk of the domain): mass transport dominates. Concentrations are nearly linear in y (steady-state diffusion under a flux boundary condition is just a linear profile). Charge density is negligible ($\sum_i z_i c_i \approx 0$, electroneutrality).

Inner region ($y \lesssim \lambda_D$, the Debye layer): screening dominates. Concentrations follow Boltzmann. Charge density is nonzero, and that's exactly what Poisson's equation balances.

Outer surface: the conceptual boundary at $y \approx \lambda_D$ where the two regions meet. Quantities with subscript o (e.g. H_o, φ_o) are evaluated here.

The IC strategy is to build each region separately and glue them together at the outer surface.

3 Step 1 — Outer linear profiles

In the outer region, only diffusion matters. At steady state with a constant flux at the electrode and Dirichlet (bulk) at $y = 1$, the profile is linear. The species fluxes equal the BV reaction rates:

$$J_{\mathrm{O_2}} = R_1 \qquad (\mathrm{O_2} \text{ consumed by } \mathrm{R_1}) \tag{3}$$

$$J_{\mathrm{H_2O_2}} = -R_1 + R_2 \qquad (\text{produced by } \mathrm{R_1}, \text{ consumed by } \mathrm{R_2}) \tag{4}$$

$$J_{\mathrm{H^+}} = R_1 + R_2 \qquad (\text{2 protons consumed in each reaction; the 2 cancels, see below}) \tag{5}$$

The corresponding linear profiles, written in terms of bulk values (O_b, P_b, H_b) and outer-surface values (O_s, P_s, H_o), are

$$\boxed{O_s = O_b - \frac{R_1}{D_O}, \qquad P_s = P_b + \frac{R_1 - R_2}{D_P}, \qquad H_o = H_b - \frac{R_1 + R_2}{D_H}.} \tag{6}$$

Why the proton denominator is D_H, not $2D_H$

The two-proton stoichiometry of each reaction would naively give $(R_1 + R_2)/(2D_H) \cdot 2$ in the numerator. But *ambipolar transport* doubles the effective proton diffusivity: because electroneutrality enforces that $\mathrm{H^+}$ and $\mathrm{ClO_4^-}$ move together, the slower partner sets the pace, and the field drags the faster one. The result is that the effective proton diffusivity is $2D_H$, not D_H, and the factor of 2 cancels against the stoichiometric 2. The bare D_H in the denominator is correct. (Comment in `forms_logc.py:798--800` explicitly warns: "do not correct".)

So the outer-region values (O_s, P_s, H_o) are determined as soon as we know the rates (R_1, R_2). But the rates depend on the surface concentrations, which depend on ψ_D, which depends on H_o...

4 Step 2 — The Picard loop

We have a self-referential algebraic system:

- Rates R_1, R_2 depend on the *surface* (inner) value $H_s = H_o \exp(-\psi_D)$.
- H_o depends on R_1, R_2 via the outer linear profile.
- ψ_D depends on φ_o, which depends on H_o.

This is solved by Picard iteration. The whole loop is closed-form algebra — no PDE solve, no Newton. Each iteration is cheap.

4.1 The 2×2 system

Inside the loop, given a current guess for (R_1, R_2):

1. Compute outer values:
$$H_o = H_b - (R_1 + R_2)/D_H,$$
and from outer-region electroneutrality plus the Boltzmann profile of the counterion,
$$\varphi_o = \ln(H_o/c^{\mathrm{bulk}}_{\mathrm{ClO_4}}), \qquad \psi_D = \varphi_{\mathrm{applied}} - \varphi_o.$$

2. Compute the surface proton concentration:
$$H_s = H_o \exp(-\psi_D).$$

3. Build BV rate constants. R_1 has two branches (cathodic "forward", anodic "backward"); R_2 here keeps only the cathodic branch:
$$\begin{aligned} A_1 &= k_1\,(H_s/H_{\mathrm{ref}})^2\,\exp(-\alpha_1 n_e \eta_1) && \mathrm{R_1\ cathodic},\\ B_1 &= k_1\,\exp\big((1-\alpha_1)\,n_e\eta_1\big) && \mathrm{R_1\ anodic},\\ A_2 &= k_2\,(H_s/H_{\mathrm{ref}})^2\,\exp(-\alpha_2 n_e \eta_2) && \mathrm{R_2\ cathodic}. \end{aligned}$$
The H_s enters squared because each reaction consumes 2 protons.

4. Solve the 2×2 system for new rates:
$$\begin{bmatrix} 1 + A_1/D_O + B_1/D_P & -B_1/D_P \\ -A_2/D_P & 1 + A_2/D_P \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \end{bmatrix} = \begin{bmatrix} A_1 O_b - B_1 P_b \\ A_2 P_b \end{bmatrix}. \tag{7}$$

5. Damped update: $R_j \leftarrow (1-\omega)R_j^{\mathrm{old}} + \omega R_j^{\mathrm{new}}$ with $\omega = 0.5$.

6. Loop until $|\Delta R_1| + |\Delta R_2| < 10^{-6}$, max 50 iterations.

4.2 Where this matrix came from

The matrix is just a rearrangement of mass balance + the linear-profile formulas. Take R_1:
$$R_1 \;=\; A_1\,O_s \;-\; B_1 P_s \;=\; A_1\,(O_b - R_1/D_O) \;-\; B_1\,(P_b + (R_1 - R_2)/D_P)$$
Collecting R_1, R_2 on the left:
$$\big(1 + A_1/D_O + B_1/D_P\big)R_1 \;-\; (B_1/D_P)R_2 \;=\; A_1 O_b - B_1 P_b.$$
Same exercise on R_2 gives the second row. So this matrix is just "write down what each rate equals, substitute the linear profile, collect terms". Nothing fancy.

4.3 Failure handling

If the Picard loop diverges, hits NaN, or hits a singular Jacobian, the IC silently **falls back to a trivial linear-φ IC**. This happens in practice below $V_{\mathrm{RHE}} = +0.20$ V, where the H^+ mass-transport limit makes the loop oscillate. The fallback is recorded in `ctx['initializer_fallback'] = True`.

5 Step 3 — Inner profile: the Debye layer

After Picard, we have $(R_1, R_2, H_o, \varphi_o, \psi_D, O_s, P_s)$ — all scalar values at the outer surface. We now need to extend each field $u_i(y)$ and $\varphi(y)$ over the *whole mesh*, not just point values.

The outer region is easy — linear profiles already given. The hard part is the inner Debye layer, where ψ goes from ψ_D at the electrode to roughly 0 at the outer surface, over a distance of λ_D.

5.1 The Gouy–Chapman profile (linear-Debye limit)

For a 1:1 symmetric electrolyte, the nonlinear Poisson–Boltzmann equation in 1D admits a closed-form solution:

$$\psi_{\mathrm{GC}}(y) \;=\; 4\,\mathrm{atanh}\Big[\tanh(\psi_D/4)\, e^{-y/\lambda_D}\Big]. \tag{8}$$

This is the *Gouy–Chapman profile*. At small ψ_D (the linear-Debye limit) it collapses to a pure exponential $\psi(y) \approx \psi_D\, e^{-y/\lambda_D}$. For larger ψ_D it stays finite but the inner ramp is steeper.

Numerical rewrite. Direct evaluation of $4\,\mathrm{atanh}$ overflows when $\psi_D \gtrsim 30$. Use the identity

$$4\,\mathrm{atanh}(x) \;=\; 2\ln\!\left(\frac{1+x}{1-x}\right),$$

which stays well-behaved as $|x| \to 1$ (just clamp the argument slightly inside ± 1). The implementation does exactly this.

5.2 When linear-Debye is not enough: Bikerman steric

Pure Boltzmann assumes ions are point charges. At the electrode, $\psi_D = +30$ would give a counterion concentration of $c_b \cdot e^{30} \approx c_b \cdot 10^{13}$ — which is physically absurd because the ions have a finite volume and can only pack so tightly.

Bikerman steric is the simplest fix: penalize the local packing fraction in the chemical potential. Ions can pack up to $1/a$ where a is the (nondimensional) ion volume. Mathematically, the modified Boltzmann distribution becomes

$$c_b(y) \;=\; c_b^{\mathrm{bulk}} \frac{\big(1 - \sum_j a_j c_j(y)\big)\; \exp(-z_b \psi(y))}{\theta_b + a_b c_b^{\mathrm{bulk}} \exp(-z_b \psi(y))}, \tag{9}$$

which caps $c_b \to 1/a_b$ as $\psi \to -\infty$. This denominator is what bounds the Poisson source above $V_{\mathrm{RHE}} = +0.66$ V where the unbounded Boltzmann co-ion would have made the Poisson residual blow up.

5.3 The BKSA composite-ψ profile

Once steric effects matter, the $\psi(y)$ profile changes character. Instead of one smooth tanh-like curve, it has *two zones*:

Two-zone Debye structure under steric saturation

Saturation zone (close to electrode): the counterion is at close-packing density $c \approx 1/a$. Charge density is roughly constant in this zone. Poisson's equation $-\varepsilon \nabla^2 \psi = \rho$ with constant

ρ gives a *linear* ψ profile (think of it like a parallel-plate capacitor field):

$$\psi(y) \;=\; \mathrm{sgn}(\psi_D)\,(|\psi_D| - \alpha y), \qquad 0 \le y < y_{\mathrm{match}}.$$

Outer Debye zone (linear-Debye regime): once enough screening has happened that $|\psi|$ drops below the saturation threshold ψ_{sat}, ions are no longer at close-packing and we return to ordinary exponential decay:

$$\psi(y) \;=\; \mathrm{sgn}(\psi_D)\,\psi_{\mathrm{sat}}\,\exp\!\big(-(y - y_{\mathrm{match}})/\lambda_D\big), \qquad y \ge y_{\mathrm{match}}.$$

The match point y_{match} and slope α come from matched asymptotics (Bazant–Kilic–Storey–Ajdari, BKSA):

$$\nu \;=\; 2\,a_{\mathrm{Cl}}\,c_{\mathrm{Cl}}^{\mathrm{bulk}} \qquad \text{(steric crowding parameter)} \tag{10}$$

$$\psi_{\mathrm{sat}} \;=\; \ln(2/\nu) \qquad \text{(potential at which counterion close-packs)} \tag{11}$$

$$\alpha \;=\; \sqrt{\frac{2}{\nu\lambda_D^2}\,\ln\!\big(1 + \nu(\cosh\psi_D - 1)\big)} \qquad \text{(slope in the saturated zone)} \tag{12}$$

$$y_{\mathrm{match}} \;=\; (|\psi_D| - \psi_{\mathrm{sat}})/\alpha \qquad \text{(thickness of the saturated zone)} \tag{13}$$

If $|\psi_D| < \psi_{\mathrm{sat}}$, the saturated zone has zero thickness and we fall back to the pure linear-Debye exponential (or the Gouy–Chapman tanh, depending on regime).

Where this comes from. The first integral of the steric Poisson–Boltzmann equation is $\frac{1}{2}\lambda_D^2(\psi')^2 \;=\; (1/\nu)\ln(1 + \nu(\cosh\psi - 1))$. Setting $\psi = \psi_{\mathrm{sat}}$ gives α as the slope at the saturation boundary, and the linear extrapolation backward to $\psi = \pm\psi_D$ gives y_{match}. The exponential outer piece is the linearization of Gouy–Chapman in the small-ψ limit. So the composite profile is just *matched asymptotics glued together at the boundary where saturation ends.*

6 Step 4 — The multispecies γ correction

Even with the right $\psi(y)$ in hand, naively writing $c_i(y) = c_i^{\mathrm{outer}}(y) \cdot \exp(-z_i\psi(y))$ **violates** the Bikerman packing constraint at the surface — the constraint and the simple Boltzmann formula are inconsistent.

The fix is a closed-form multiplicative correction $\gamma(y)$ that restores consistency. From multispecies zero-flux equilibrium with the steric chemical potential, one derives

$$\boxed{\gamma(y) \;=\; \frac{1}{1 + \sum_j a_j\,c_j^{\mathrm{anchor}}(y)\,\big(\exp(-z_j\psi(y)) - 1\big)}.} \tag{14}$$

Two convenient features of this expression:

- **Neutral species drop out** automatically, because $e^0 - 1 = 0$. Only charged species (H^+ and ClO_4^-) contribute to the denominator.
- In the linear-Debye limit (ψ small), $e^{-z\psi} - 1 \approx -z\psi$ and $\gamma \to 1$ smoothly. So the correction vanishes in the regime where it shouldn't matter.

The IC then seeds every species on the saturated steric manifold:

$$u_{O_2}^{IC}(y) = \ln O_{outer}(y) + \ln \gamma(y), \tag{15}$$
$$u_{H_2O_2}^{IC}(y) = \ln P_{outer}(y) + \ln \gamma(y), \tag{16}$$
$$u_{H^+}^{IC}(y) = \ln H_{outer}(y) - \psi(y) + \ln \gamma(y), \tag{17}$$
$$\varphi^{IC}(y) = \ln\big(H_{outer}(y)/c_{ClO_4}^{bulk}\big) + \psi(y). \tag{18}$$

What the pieces are doing, in plain English

- "ln outer" : the bulk-to-outer-surface diffusion ramp.
- "$-\psi(y)$" on H^+ : Boltzmann depletion of the proton in the Debye layer (a +1 ion is repelled from a positive electrode, so its concentration drops).
- "$+\psi(y)$" on φ : the actual electrostatic potential profile, going from the bulk-electroneutrality reference up to $\varphi_{applied}$ at the electrode.
- "$+ \ln \gamma$" on every species : the steric correction that makes the Boltzmann profile obey the packing constraint.

7 Step 5 — Why μ_H is the Right primary variable

The production solver does *not* carry $u_H = \ln c_H$ as the proton's primary unknown. It carries the **electrochemical potential**

$$\mu_H \;=\; u_H + z_H\varphi \;=\; u_H + \varphi \qquad (z_H = +1).$$

Concentrations are reconstructed as $c_H = \exp(\mu_H - \varphi)$.

7.1 Why this helps

Inside the Debye layer where H^+ is in local equilibrium, $\nabla\mu_H \approx 0$. So μ_H is *nearly constant in y across the Debye layer* — exactly where u_H and φ are each varying by tens of log-units. Newton sees a smooth unknown in the regime where the alternatives are wildest.

7.2 Exact cancellation under the Debye–Boltzmann IC

A clean correctness check: under the IC of Section 4 (ignoring γ for clarity),

$$\begin{aligned}\mu_H^{IC}(y) &= u_H^{IC}(y) + \varphi^{IC}(y)\\ &= \big[\ln H_{outer}(y) - \psi(y)\big] + \big[\ln\big(H_{outer}(y)/c_{Cl}^{bulk}\big) + \psi(y)\big]\\ &= 2\ln H_{outer}(y) - \ln c_{Cl}^{bulk}.\end{aligned}$$

The diffuse-layer ψ — the wildest-varying piece — cancels exactly. The IC for μ_H contains *only* the slowly-varying outer profile of H^+ and a constant offset. That is as smooth as it gets.

8 Putting it all together

8.1 Algorithm summary

1. Read bulk values, diffusivities, BV constants, and steric sizes from `solver_params`.

2. Picard-iterate the algebraic 2×2 system to get (R_1, R_2), (O_s, P_s, H_o), and ψ_D at the outer surface. Bail to linear-φ on failure.

3. Build the **outer linear profiles** $O_{\text{outer}}(y)$, $P_{\text{outer}}(y)$, $H_{\text{outer}}(y)$ as Firedrake expressions on the mesh.

4. Build the **inner** $\psi(y)$ **profile**: pure Gouy–Chapman if the electrolyte is not steric, BKSA composite if it is.

5. Build the **multispecies** $\gamma(y)$ **correction** (when steric).

6. Interpolate $u_i^{\text{IC}}(y)$ and $\varphi^{\text{IC}}(y)$ onto the mesh; copy into `ctx['U']` and `ctx['U_prev']`.

7. Hand to Newton.

8.2 What it buys you, in numbers

Metric	Linear-φ IC	Debye–Boltzmann IC
Newton iters at $V_{\text{RHE}} = +0.30$ V	~ 50	~ 10–30 (1.7–6.3× fewer)
Cold-start ceiling	+0.30 V	+0.60 V
Warm-walk reach (with Stern)	+0.60 V	+1.00 V (15/15)

The Bikerman-consistent IC alone (composite-ψ + multispecies-γ, "Option 2b") accounts for the +0.4 V gain from +0.60 to +1.00 V on the production stack.

8.3 Soft spots to be aware of

- **Picard can fail silently.** The fallback to linear-φ is automatic but is a no-op below $V_{\text{RHE}} =$ +0.20 V (the H^+ mass-transport limit makes the outer loop oscillate).

- **The IC is not on the adjoint tape.** The whole body runs under `firedrake.adjoint.stop_annotating()`. The IC is an initial *guess*, not a control — you don't want pyadjoint to think the Picard iterations are part of the gradient computation.

- **It is still a guess.** If the production-stack residual changes (e.g. a different counterion model, a different BV formulation), the IC can become inconsistent with what Newton is converging to, and convergence degrades. The IC and the residual must agree about what the saturated steric manifold looks like; this is hard rule #3 in `CLAUDE.md`.

9 One-paragraph explainer for the meeting

The version to say out loud

The Debye–Boltzmann IC is an analytical sketch of the steady-state solution that we hand to Newton as a starting guess. It builds the solution in two pieces: an outer linear-diffusion profile from bulk to a fictitious "outer surface" just outside the Debye layer, and an inner Boltzmann-style screening profile from the outer surface down to the electrode. The two are stitched together using a self-consistent algebraic 2×2 Picard solve for the BV reaction rates, a Gouy–Chapman or BKSA composite $\psi(y)$ profile inside the Debye layer, and a multispecies activity correction $\gamma(y)$ that keeps the Bikerman packing constraint satisfied. With the IC, Newton converges 1.7–6.3× faster, and unblocks cold starts that the linear IC could not reach. It is not part of the residual and not on the adjoint tape — it is a guess, optimized to be right almost everywhere.